

Paul Iacobucci

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Education

Cornell University

Expected May 2027

Bachelor's in Computer Science; Minors in Artificial Intelligence, Applied Economics (Dyson)

GPA: 3.64

Relevant Coursework: Machine Learning, Natural Language Processing, AI Reasoning, Systems Programming, Computer Networks, Embedded Systems, Robotics, Data Structures & Algorithms, Functional Programming

Experience

L3Harris Technologies

Rochester, NY

Machine Learning Engineer Intern

May 2026 – Aug. 2026

- Built a real-time C++ text-to-speech AI system running fully on-device on an embedded military radio.
- Quantized a 15M-param VITS model Float32→INT8 via ARM NEON SIMD, cutting storage 70% to 17.79MB.
- Tuned ONNX Runtime memory arenas and stripped eSpeak NG to under 5MB, keeping active RAM below 45MB.
- Built a student-model training and ASR quality-gate pipeline, rejecting a faster model that failed intelligibility.

Software Engineering Intern

Dec. 2025 – Jan. 2026

- Engineered a Java streaming plugin for Android defense hardware, cutting live ATAK video feed latency by 20%.
- Offloaded real-time signal parsing from CPU to FPGA fabric, reducing system-wide overhead by 40%.

Software Engineering Intern

May 2025 – Aug. 2025

- Developed a custom mapping and video streaming application for Linux-based military hardware.
- Extended a Go (Fiber) microservice sustaining 1,000+ concurrent streams and WebSocket chats using Goroutines.
- Achieved sub-1s video latency using MediaMTX to transmux WebRTC/RTSP/HLS for TCP/UDP connections.

Cornell Zhang Research Group

Ithaca, NY

Machine Learning Research Assistant

Jan. 2026 – Present

- Profiled frontier AI models on 8x H100 GPU clusters for an industry client designing custom AI silicon.
- Diagnosed expert-parallel bottlenecks via SGLang and Nsight, revealing 33% disparity and 117 μ s waits.
- Extended LLMservingSim with a per-layer Roofline model to classify compute- and memory-bound layers.
- Proved EP penalties scale strictly with intermediate dimension sizes via TP/EP scaling sweeps.
- Analyzing KV-cache behavior in LLM RL post-training to evaluate a proposed 6TB, 900GB/s memory tier.

Projects

Haven: Wearable Hearing-Protection Device ↗

2026

- Building a hearing-protection wearable (nRF5340 + ADAU1860 DSP) with a React Native BLE tuning app.
- Took an OpenEarable-based PCB port to a fabrication-ready 6-layer board with a 219-item DRC baseline.
- Designed a safety-critical hearing test with a hard-capped 85dB ceiling, watchdog timeout, and instant-stop failsafes.

Scout (Backed by Cornell) ↗

2025 – 2026

- Built a full-stack student-athlete platform backed by a \$6K Cornell grant, scaling to 200+ active users.
- Architected a Next.js 14 and Tailwind frontend backed by Supabase PostgreSQL with Row Level Security.
- Integrated the Claude API for personalized AI outreach and deployed via Vercel with GitHub Actions CI/CD.

Triton GPU Performance Kernels (NVIDIA H100) ↗

Apr. 2026

- Wrote custom GPU kernels accelerating core AI model operations beyond native PyTorch performance.
- Fused FP8 quantization and LayerNorm on H100 SXM, hitting 3,531.8 GB/s effective bandwidth (45.7% speedup).

Mini-TensorRT: Deep Learning Graph Compiler ↗

Mar. 2026

- Built a from-scratch, zero-dependency C++ compiler that optimizes trained AI models for faster inference.
- Fused Conv-ReLU kernels to cut DRAM round-trips, measuring 11.2% speedup on cache-exceeding inputs.

Technical Skills

Languages: C++, Rust, Python, C, SystemVerilog, Go, Java, SQL, Bash

Hardware & Compute: CUDA, Triton, Vitis HLS, FPGA, PyTorch, TensorRT, ONNX Runtime, NCCL

Tools & Systems: Linux, Docker, Yocto/BitBake, Nsight, FastAPI, GitHub Actions, AWS

Interests: Poker, Chess, Powerlifting (315 lb bench), Golf